Response to Referee #5

We thank the referee for a very helpful and detailed report. The comments have substantially improved the clarity, transparency, and pedagogical structure of the paper. Below we respond to each point in turn. For ease of reference, referee remarks appear in italics.

1. Two-agent structure and literature positioning

Referee comment:

> *The paper presents the two-agent model as if it were being used for the first time. It would be preferable to acknowledge and cite the earlier literature that introduced and developed this framework.*

Response:
We thank the referee for this suggestion. The two-agent (Ricardian vs. hand-to-mouth) household structure is well established in the DSGE literature, and we agree that the original manuscript did not sufficiently emphasize these precedents.

To address this point, we now explicitly acknowledge the literature using rule-of-thumb or hand-to-mouth households in DSGE models, including Coenen and Straub (2005) and Galí, López-Salido and Vallés (2007).

We clarified the literature positioning in the model section when introducing households in footnote 3:

> This specification follows the standard two-agent structure widely used in DSGE models with fiscal policy (e.g. Coenen and Straub, 2005; Galí, López-Salido and Vallés, 2007).

These additions clarify that the two-agent structure used in the paper follows an established modeling approach.

2. Hand-to-mouth households: optimization problem and FOCs

Referee comment:

> *The static optimization problem of the hand-to-mouth households is not specified, and the corresponding optimality conditions for their consumption and labor choices are absent.*

Response:
We thank the referee for pointing this out. In the revised manuscript we now explicitly state the static optimization problem of the hand-to-mouth households and report their intratemporal optimality condition linking consumption and labor supply in the appendix.

In the model section, we now introduce the problem faced by these households, on page 8:

> The subset of households that do not have access to financial assets or capital accumulation receive only labor income and government transfers, TR_t^{hm}, which they use to consume and pay the consumption tax. We assume that hand-to-mouth households do not pay labor income taxes, reflecting either thresholds in the tax code or their representation of informal sector workers. Hence, they face the following simpler budget constraint:
>
> $$(1+\tau_t^c)p_t C_t^{hm} = W_t h_t^{hm} + TR_t^{hm}. \tag{1}$$
>
> They solve a static optimization problem, choosing C_t^{hm} and h_t^{hm} to maximize (1), subject to (6).

The full derivation of the first-order conditions is in Appendix A (equations A.2 and A.5).

3. Determinants of the government interest rate

Referee comment:

> *It would be useful for the authors to elaborate on the determinants of the interest rate paid by the government and whether it is linked to the rate faced by Ricardian households.*

Response:

We thank the referee for this suggestion. In the model, both the government and Ricardian households borrow from international financial markets and therefore face the same external interest rate. This rate is given by the world interest rate plus a sovereign risk premium that depends on the economy's total external-debt-to-GDP ratio. As a result, the model assumes no segmentation between public and private external borrowing.

To clarify this point, we added footnote 8 in the model section, as follows:

> Both the government and Ricardian households borrow at the same external interest rate, given by the world interest rate plus a risk premium that depends on the economy's total external-debt-to-GDP ratio. This specification implies that changes in external indebtedness affect the borrowing costs of both the public and private sectors.

4. Dynamics of foreign GDP

Referee comment:

> *The paper would benefit from a clearer exposition of the dynamics of foreign GDP and whether it is related to the non-stationary productivity shock.*

Response:

We thank the referee for this suggestion. In the model, detrended foreign GDP is specified as an exogenous stochastic process that captures fluctuations in external demand faced by the domestic economy. By assumption, it is independent of the domestic non-stationary productivity shock, which governs the long-run growth dynamics of non-oil output in the model.

To clarify this point, we added the following footnote 15:

> Detrended foreign output is modeled as an exogenous AR(1) process that captures fluctuations in external demand. It is independent of the domestic productivity process that drives the trend growth of the economy. This assumption reflects the small open economy perspective adopted in the model, whereby domestic developments do not influence foreign output.

5. First-order conditions and model transparency

Referee comment:

> *The paper frequently presents optimization problems without deriving or discussing the corresponding first-order conditions.*

Response:

We thank the referee for this comment. The full set of stationary equilibrium conditions, including the first-order conditions associated with the optimization problems presented in the model section, was already reported in Appendix A. However, we agree that the connection between the optimization problems described in the main text and the corresponding equilibrium conditions in the Appendix could be made clearer.

To address this point, we now explicitly guide the reader to the Appendix when presenting the model. In particular, we clarified the sentence that introduces the equilibrium conditions with the following text of footnote 16:

> The full set of stationary equilibrium conditions, including the first-order conditions associated with the optimization problems described above, is reported in Appendix A.

This clarification makes it easier for readers to locate the optimality conditions without interrupting the exposition of the model in the main text. We do not believe we need to interpret these optimality conditions in this paper as they are standard in this literature.

6. Equation numbering

Response:

We have implemented consistent equation numbering throughout the manuscript for ease of reference. If we don't refer to an equation in the text, we are not numbering it.

7. Transmission channels of commodity revenue

Referee comment:

> *The paper should more clearly outline the mechanisms through which revenue from the non-produced commodity affects the domestic economy.*

Response:

We thank the referee for this suggestion. The transmission mechanisms of commodity revenue are already embedded in the model and illustrated in the subsection *"Effects of an oil price shock"*, which presents impulse responses to an oil-price disturbance. However, we agree that these mechanisms were not sufficiently summarized in a single place.

To address this point, we added a paragraph at the end of that subsection to clearly outline the transmission channels. The following text was included on page 18:

> In summary, an increase in oil prices raises government revenue, improving the fiscal balance and reducing the need for external borrowing. Lower external debt leads to a decline in the sovereign risk premium, which reduces the domestic interest rate faced by both the government and private agents. This stimulates private investment and consumption. Through these channels, oil-price movements propagate to the external and fiscal balances and, ultimately, to output

This addition makes explicit how commodity-revenue fluctuations are transmitted through fiscal policy, external borrowing conditions, and private-sector behavior.

8. Mechanisms underlying fiscal consolidation

Referee comment:

> *The analysis in the fiscal consolidation section insufficiently clarifies how fiscal instruments interact with household behavior, country risk dynamics, and commodity-price shocks.*

Response:

We thank the referee for this comment. The mechanisms underlying fiscal consolidation are embedded in the model and discussed in Section 6, but we agree that the exposition can be made clearer.

To address this point, we added a paragraph at the beginning of the fiscal consolidation section to provide a structured description of the key transmission channels. The following text was included on page 22:

> A high-level description of the effects of fiscal consolidation is as follows (a detailed discussion appears below). The effects arise from the interaction between household decisions, production, and financial conditions. Fiscal instruments affect households through their budget constraints and intratemporal incentives: higher labor taxes distort labor supply by reducing after-tax wages, while transfers directly affect the income and labor-supply choices of hand-to-mouth households. Public consumption influences household utility, particularly when it is complementary to private consumption, thereby affecting consumption–labor trade-offs. Public investment operates through the accumulation of public capital, which enters firms' production functions and affects the marginal productivity of private inputs. At the same time, fiscal consolidation reduces the need for external borrowing, putting downward pressure on the sovereign risk premium. The resulting decline in borrowing costs lowers the user cost of capital and

> stimulates private investment. These effects are heterogeneous across households: hand-to-mouth households respond primarily to contemporaneous income changes, while Ricardian households adjust consumption, labor supply, and investment intertemporally. The overall dynamics of the economy reflect the interaction between labor-supply adjustments, the evolution of private and public capital, and the financial channel operating through the risk premium along the transition path.

This addition clarifies how fiscal instruments, household behavior, production, and country risk jointly shape the transition dynamics under fiscal consolidation.

9. Role of the risk premium in investment dynamics

Referee comment:

> *The simulations show a substantial response of private investment to changes in the premium, but the mechanism is not clearly articulated.*

Response:

We thank the referee for this comment. The role of the sovereign risk premium in shaping private investment is a central mechanism in the model and is illustrated in the subsection *"Effects of a country premium shock"*, where an increase in the premium raises borrowing costs and leads to a decline in private investment.

We agree, however, that this mechanism was not sufficiently connected to the fiscal consolidation analysis. To address this point, we added a paragraph in the fiscal consolidation section clarifying how the risk premium affects investment decisions along the transition path. The following text was included on page 25:

> The decline in the deficit (panel 2-3) contains public debt, but private debt (not shown) increases to finance private consumption and investment, which initially puts upward pressure on the premium. The eventual decline in overall debt leads to a decline of the country risk premium, reaching about 35 bps below steady state by the end of the consolidation horizon (panel 2-4), which in turn increases private investment (panel 3-3) because of the lower cost of capital.

We also note that while the impulse-response analysis we presented in the sub-section "Effects of a country premium shock" considers an unexpected increase in the risk premium, the fiscal consolidation exercise is conducted under perfect foresight, where agents anticipate the evolution of fiscal policy and debt. Despite this difference in timing, the underlying transmission mechanism—through borrowing costs and the user cost of capital—remains the same.

10. Joint demand- and supply-side effects of consolidation

Referee comment:

> *Fiscal consolidation simultaneously affects aggregate demand and supply, but this is not clearly explained.*

Response:

We thank the referee for this comment. We agree that fiscal consolidation affects multiple margins of adjustment, but we would like to clarify that, in our framework, these effects are best understood through the lens of a real model with flexible prices rather than through a demand–supply decomposition.

In the model, output is determined by the combination of aggregate hours worked, private capital, and public capital. As a result, fiscal consolidation affects economic activity insofar as it changes labor-supply decisions and the accumulation of private and public capital. In the short run, adjustments in after-tax wages and disposable income lead to changes in labor supply, while over time, fiscal policy influences the paths of private investment and public capital accumulation. In addition, the reduction in public debt lowers the sovereign risk premium, which affects the user cost of capital and thereby investment decisions.

For this reason, we have framed the discussion of the consolidation in terms of real and financial channels—namely labor-supply adjustments, capital accumulation, and the risk-premium mechanism—rather than in terms of aggregate demand. The paragraph added in response to Comment 12 below provides a structured description of how each fiscal instrument operates through these channels and how their interaction shapes the transition dynamics.

We hope this clarification addresses the referee's concern.

11. Labor-supply responses

Referee comment:

> *The reasons behind shifts in hours worked are not well articulated.*

Response:

We thank the referee for this comment. The model features two types of households whose labor-supply decisions respond to different mechanisms, and we agree that this distinction should be explained more clearly.

To address this point, we added a paragraph in the fiscal consolidation section clarifying the determinants of household decisions for each household type. The following text was included on page 26:

> Labor-supply responses differ across households due to their access to financial markets. Ricardian households adjust hours worked based on both intratemporal substitution (changes in after-tax wages) and intertemporal considerations, as they can smooth consumption over time. Hand-to-mouth households, in contrast, do not participate in financial markets and therefore choose labor supply based only on current income conditions, responding to intratemporal incentives. Both types of households are also affected by wealth effects: higher income or transfers tend to reduce labor supply, while lower income increases it. As a result, fiscal consolidation—through higher labor taxes and lower transfers—tends to increase labor supply, particularly among hand-to-mouth households (panel 4-4), while Ricardian households exhibit a less pronounced response due to their intertemporal optimization (panel 4-3).

This addition clarifies how labor supply is determined in the model and how different households respond to fiscal policy changes.

12. Interaction and relative contribution of fiscal instruments

Referee comment:

> *The consolidation package has multiple components; the reader would benefit from a discussion of their relative contributions.*

Response:

We thank the referee for this suggestion. The fiscal consolidation program combines multiple instruments, and we agree that clarifying their roles improves the interpretation of the results. While the paper does not perform a full quantitative decomposition of each instrument, the mechanisms can be understood by examining how each component affects the economy within the model.

To address this point, we added a paragraph in the fiscal consolidation section summarizing the role of each fiscal instrument. The following text was included on page 27:

> To summarize this section: The consolidation package combines several fiscal instruments that affect the economy through real and financial channels. Increases in labor-income taxes distort labor supply decisions by reducing after-tax wages and disposable income. Changes in capital taxation affect investment decisions through their impact on the user cost of capital. Reductions in transfers directly affect hand-to-mouth households, altering their income and labor-supply choices. Cuts in public consumption affect household utility and intratemporal allocations, without directly entering production. Reductions in public investment have more persistent effects, as they slow the accumulation of public capital and thus reduce future productive capacity. In addition to these direct effects, fiscal consolidation reduces the need for external public borrowing, putting downward pressure on the sovereign risk premium. The resulting decline in borrowing costs lowers the user cost of capital and stimulates private investment, supporting the accumulation of private capital. The overall dynamics of the consolidation reflect the interaction between labor-supply adjustments, capital accumulation—both private and public—and the financial channel operating through the risk premium. Appendix H illustrates the effects of adjusting each fiscal instrument separately to achieve a fiscal deficit reduction of one percent of GDP.

13. U-shaped response of aggregate consumption

Referee comment:

> *The explanation of the U-shaped consumption response should be expanded.*

Response:

We thank the referee for this suggestion. The inverse U-shaped response of aggregate consumption reflects the interaction between short-run income effects and medium-run financial and investment dynamics, as well as differences across household types.

To clarify this mechanism, we added a paragraph in the fiscal consolidation section. The following text was included on page 25:

> Higher income taxes reduce the consumption of Ricardian households (panel 4-1), whereas the consumption of restricted, hand-to-mouth households (panel 4-2) follows the inverse U-shape pattern of government transfers, as this group of consumers receives the entirety of transfers. The inverse U-shaped response of aggregate consumption (panel 3-2) reflects the interaction of short-run and medium-run effects. In the short run, higher labor taxes and lower transfers reduce disposable income, leading to an overall decline in consumption. At the same time, lower public expenditure reduces overall absorption. Over time, the fiscal consolidation reduces external debt and lowers the sovereign risk premium, which decreases borrowing costs and stimulates private investment and the consumption of Ricardian households. As capital accumulates and income rises, consumption gradually recovers, especially for liquidity-constrained households. Ricardian households also smooth consumption intertemporally, contributing to the recovery.

This addition clarifies the two-stage dynamics underlying the consumption response and highlights the role of household heterogeneity and financial conditions.

14. Trade balance dynamics

Referee comment:

> *The discussion of the trade balance should better link the model's predictions to underlying drivers.*

Response:

We thank the referee for this comment. The dynamics of the trade balance follow directly from the model's national income identity, where the trade balance equals export revenues (including oil) minus imports, and imports are driven by domestic absorption.

To clarify this mechanism, we added the following footnote 32 in the fiscal consolidation section:

> The trajectory of the trade balance (panel 3-4) is largely driven by the path of oil revenues. Changes in domestic absorption—reflecting the interaction between private and public consumption and investment—also affect imports, but their net contribution is more muted relative to the direct effect of oil revenues.

This addition links the trade balance dynamics directly to the model structure and clarifies the roles of oil revenue and domestic absorption.

Concluding remark

We appreciate the referee's thorough and insightful comments. In response, we have clarified the model structure and its transmission mechanisms, strengthened the exposition of the fiscal consolidation analysis, and improved the discussion of how fiscal instruments, household behavior, and the sovereign risk premium interact. We have also refined the presentation of the simulation results, including the implementation of the consolidation under perfect foresight. These revisions improve the clarity, internal consistency, and interpretability of the paper.

Sincerely,
Manuel González-Astudillo, Juan Guerra-Salas, Avi Lipton

References

Coenen, Günter and Roland Straub. (2005) "Does government spending crowd in private consumption? Theory and empirical evidence for the euro area." *International finance*, 8, 435–470.

Galí, Jordi, J David López-Salido, and Javier Vallés. (2007) "Understanding the effects of government spending on consumption." *Journal of the european economic association*, 5, 227–270.